# RASHI VERMA

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## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2027 | 9.00 |
| DAV, Public School, Ghandhi Nagar, Ranchi, Jharkhand | Class XII (C.B.S.E) | 2014 | 82.50% |
| St. Joseph's Convent High School, Patna, Bihar | Class X (I.C.S.E) | 2012 | 86.86% |

## SCHOLASTIC ACHIEVEMENTS

Prime Minister's Research Fellowship (PMRF)

Best Oral Presentation Award at the ICAER 2025 conference, IIT BOMBAY

Attended IEEE WCCI 2026 - World Congress on Computational Intelligence, Maastricht, Netherland

## POSITIONS OF RESPONSIBILITY

- Placement Coordinator | Office of Career Services (OCS), IIT Hyderabad
- **Teaching Assistant (TA)** for *Basics of AI and ML in Chemical Engineering* and *Optimization Techniques* at **IIT Hyderabad**, supporting tutorials, hands-on sessions, and student learning.
- **Tutorial & Workshop Instructor** for **MATLAB**, conducting practical sessions and guiding students in computational and engineering applications at **BVRIT College**.

## SKILLS

Python | MATLAB | PyBaMM | PyTorch | TensorFlow / Keras | pymoo | Pyomo

## RELEVANT COURSES

Basics & Application of AI/ML for Process System Engineering

Electrochemical Energy Storage Systems: Batteries, Fuel Cells, and Supercapacitors

Advanced Numerical Methods

Molecular Modeling of Catalytic Reactions

## PROJECTS

**Explainable AI for Wind Turbine Time-Series Modeling**

*Application of TimeSHAP for Interpreting Time-Series Models of Wind Turbines*

- Applied **TimeSHAP**, an explainable AI (XAI) framework, to interpret time-series machine learning models for wind turbine data.
- Analyzed the **temporal contribution of input features** to model predictions and identified influential time steps in turbine behavior.
- Developed interpretable insights into model predictions to improve **transparency, reliability, and understanding of wind turbine time-series forecasting**.

**Multi-Objective Optimization of Lithium-Ion Battery Charging Protocols**

- Formulated a **multi-objective optimization framework** for designing fast and safe charging protocols for lithium-ion

batteries.
- Applied **Bayesian Optimization and NSGA-II** to optimize charging parameters considering trade-offs among **charging time, thermal behavior, and charge capacity**.
- Developed and evaluated **Pareto-optimal multi-stage constant-current (MSCC) charging strategies** subject to voltage, temperature, and lithium-plating constraints.

**Surrogate Model-Assisted Warm Starting for Water Network Optimization**

- Developed an **ANN-based surrogate model** using 5,000 samples to capture objective and constraint behavior for a **79-variable MINLP water network optimization** problem.
- Integrated the surrogate with **NSGA-II** to generate Pareto-optimal solutions by minimizing the objective function and constraint violations.
- Used the resulting solutions as **warm-start initializations for the MINLP solver**, improving initial solution quality and accelerating solver convergence.

## EXTRACURRICULAR

Badminton

## PUBLICATIONS

- Bayesian Optimization Framework for Multi-Objective Charging Strategy Design in Lithium-Ion Batteries. IEEE World Congress on Computational Intelligence (WCCI) 2026.
- Optimal Multi-Stage Constant Current Charging Strategy for Lithium-ion Battery Adopting Multi-Objective Approach. Advances in Energy Research, Volume I: Select Proceedings of ICAER 2025.
- Masampally, V.S., Verma, R., Mitra, K. (2025). Application of TimeSHAP, an eXplainable AI Tool, for Interpreting Time-Series Model of Wind Turbines. In Optimization, Uncertainty and Machine Learning in Wind Energy Conversion Systems. Springer, Singapore.
- Optimization of Organosolv Process for Isolation of Lignin from Corncob in THF-H2O Solvent Mixture. Biomass Conversion and Biorefinery, 2025.
- Verma, R. & Besta, C.S. (2024). Modelling of chemical processes using artificial neural network. Indian Chemical Engineer, 66:1, 84-105.
- Verma, R., Yerolla, R., Suhailam, P., Besta, C.S. (2023). Application of Artificial Intelligence (AI) and the Internet of Things (IoT) in process industries towards Industry 4.0. In Internet of Things in Modern Computing: Theory and Applications (1st ed., pp. 12-36). CRC Press.

## AREA OF EXPERTISE

Battery Modelling | AI/ML | Multi-Objective Optimization